Yuzhou Gu

sevenkplus.g@gmail.com
https://sevenkplus.com/

WORK EXPERIENCE

OpenAI — San Francisco, CA, USA
- Member of Technical Staff, Dec. 2025 - Present

New York University — New York, NY, USA
- Data Science Faculty Fellow & Courant Instructor, Sept. 2024 - Nov. 2025

Institute for Advanced Study — Princeton, NJ, USA
- Member, Sept. 2023 - July 2024

EDUCATION

Massachusetts Institute of Technology — Cambridge, MA, USA
- Ph.D. in Electrical Engineering and Computer Science, Sept. 2018 - June 2023
  Thesis: Channel Comparison Methods and Statistical Problems on Graphs
  Advisor: Yury Polyanskiy
- M.Eng. in Electrical Engineering and Computer Science, Sept. 2017 - June 2018
  Advisor: Yury Polyanskiy
- B.S. in Electrical Engineering and Computer Science
  B.S. in Mathematics, Sept. 2013 - June 2017

INTERNSHIPS

D. E. Shaw & Co.
- Quantitative Research Intern, June - Aug. 2025 — London, UK
- Quantitative Research Intern, June - Aug. 2021 — Remote (New York, NY, USA)

Pony.ai — Fremont, CA, USA
- Engineering Intern, June - Aug. 2019
- Engineering Intern, June - Aug. 2018

Google — Mountain View, CA, USA
- Engineering Intern, May - Aug. 2015

AWARDS

- George M. Sprowls PhD Thesis Award, MIT EECS Department, 2024
- Best Student Paper Award at *Conference on Learning Theory (COLT) 2021*
- Jacobs Presidential Fellowship, MIT, 2018-2019
- Gold Medal (3rd place) at *International Olympiad in Informatics (IOI) 2012*

PUBLICATIONS & PREPRINTS

1. **Yuzhou Gu**, Mark Sellke. "A Counterexample to the Gaussian Completely Monotone Conjecture," *Preprint. arXiv:2605.11656*
2. **Yuzhou Gu**, Xin Li, Yinzhan Xu. "A Unified Lower Bound on the Noisy Query Complexity of Boolean Functions," *Conference on Learning Theory (COLT) 2026.*
3. MohammadHossein Bateni, Vincent Cohen-Addad, **Yuzhou Gu**, Silvio Lattanzi, Simon Meierhans, Christopher Mohri. "Algorithmic Thinking Theory," *Conference on Learning Theory (COLT) 2026.*

4. **Yuzhou Gu**, Yanjun Han, Jian Qian. "Evolution of Information in Interactive Decision Making: A Case Study for Multi-Armed Bandits," *Conference on Neural Information Processing Systems (NeurIPS) 2025.*
5. **Yuzhou Gu**, Xin Li, Yinzhan Xu. "Tight Bounds for Noisy Computation of High-Influence Functions, Connectivity, and Threshold," *Conference on Learning Theory (COLT) 2025.*
6. Pietro Caputo, Zongchen Chen, **Yuzhou Gu**, Yury Polyanskiy. "Entropy Contractions in Markov Chains: Half-Step, Full-Step and Continuous-Time," *Electronic Journal of Probability, 30:1-29, 2025.*
7. **Yuzhou Gu**. "Exact condensation thresholds for NAE-SAT and the hypergraph stochastic block model," *In preparation.*
8. **Yuzhou Gu**, Yury Polyanskiy. "Robust reconstruction on trees revisited," *In preparation.*
9. **Yuzhou Gu**, Zhao Song, Junze Yin. Binary Hypothesis Testing for Softmax Models and Leverage Score Models. *International Conference on Machine Learning (ICML) 2025.*
10. **Yuzhou Gu**, Nikki Lijing Kuang, Yi-An Ma, Zhao Song, Lichen Zhang. "Log-concave Sampling over a Convex Body with a Barrier: a Robust and Unified Dikin Walk," *Conference on Neural Information Processing Systems (NeurIPS) 2024.*
11. **Yuzhou Gu**, Aaradhya Pandey. "Community detection in the hypergraph stochastic block model and reconstruction on hypertrees," *Conference on Learning Theory (COLT) 2024.*
12. **Yuzhou Gu**, Yury Polyanskiy. "Non-linear Log-Sobolev Inequalities for the Potts Semigroup and Applications to Reconstruction Problems," *Communications in Mathematical Physics, 404(2):769–831, 2023.*
13. **Yuzhou Gu**, Ziqi Zhou, Onur Günlü, Rafael G. L. D'Oliveira, Parastoo Sadeghi, Muriel Médard, Rafael F. Schaefer. "Generalized Rainbow Differential Privacy," *Journal of Privacy and Confidentiality, 14(2), 2024.*
14. Zongchen Chen, **Yuzhou Gu**. "Fast Sampling of b-Matchings and b-Edge Covers," *ACM-SIAM Symposium on Discrete Algorithms (SODA) 2024.*
15. **Yuzhou Gu**, Yury Polyanskiy. "Weak Recovery Threshold for the Hypergraph Stochastic Block Model," *Conference on Learning Theory (COLT) 2023.*
16. **Yuzhou Gu**, Yury Polyanskiy. "Uniqueness of BP fixed point for the Potts model and applications to community detection," *Conference on Learning Theory (COLT) 2023.*
17. **Yuzhou Gu**, Zhao Song, Lichen Zhang. "Faster Algorithms for Structured Linear and Kernel Support Vector Machines," *International Conference on Learning Representations (ICLR) 2025.*
18. **Yuzhou Gu**, Zhao Song, Junze Yin, Lichen Zhang. "Low Rank Matrix Completion via Robust Alternating Minimization in Nearly Linear Time," *International Conference on Learning Representations (ICLR) 2024.*
19. **Yuzhou Gu**, Zhao Song. "A Faster Small Treewidth SDP Solver," *Preprint. arXiv:2211.06033*
20. **Yuzhou Gu**, Yinzhan Xu. "Optimal Bounds for Noisy Sorting," *ACM Symposium on Theory of Computing (STOC) 2023.*
21. Emmanuel Abbe, Elisabetta Cornacchia, **Yuzhou Gu**, Yury Polyanskiy. "Stochastic block model entropy and broadcasting on trees with survey," *Conference on Learning Theory (COLT) 2021.* **Best Student Paper Award.**

22. **Yuzhou Gu**, Adam Polak, Virginia Vassilevska Williams, Yinzhan Xu. "Faster monotone min-plus product, range mode, and single source replacement paths," *International Colloquium on Automata, Languages and Programming (ICALP) 2021.*
23. **Yuzhou Gu**, Hajir Roozbehani, Yury Polyanskiy. "Broadcasting on trees near criticality," *IEEE International Symposium on Information Theory (ISIT) 2020.*
24. Zeev Dvir, Sivakanth Gopi, **Yuzhou Gu**, Avi Wigderson. "Spanoids - an abstraction of spanning structures, and a barrier for LCCs," *Innovations in Theoretical Computer Science (ITCS) 2019.*
    Zeev Dvir, Sivakanth Gopi, **Yuzhou Gu**, Avi Wigderson. "Spanoids - an abstraction of spanning structures, and a barrier for LCCs," *SIAM Journal on Computing (SICOMP), 49(3):465–496, 2020.*
25. Lijie Chen, Erik D. Demaine, **Yuzhou Gu**, Virginia Vassilevska Williams, Yinzhan Xu, Yuancheng Yu. "Nearly optimal separation between partially and fully retroactive data structures," *Scandinavian Symposium and Workshops on Algorithm Theory (SWAT) 2018.*
26. **Yuzhou Gu**. "Zero-error communication over adder MAC," *Preprint. arXiv:1809.07364*
27. **Yuzhou Gu**. "Graph magnitude homology via algebraic Morse theory," *Preprint. arXiv:1809.07240*
28. **Yuzhou Gu**. "Generalized equivariant model structure on $\mathbf{Cat}^{I}$," *Preprint. arXiv:1605.07983*
29. **Yuzhou Gu**. "Some results on reversible gate classes over non-binary alphabets," *Preprint. arXiv:1606.00804*

TALKS

1. Reconstruction on hypertrees, *Penn/Temple Probability Seminar.* University of Pennsylvania, Philadelphia, PA, Oct. 2024.
2. Community detection in the hypergraph stochastic block model, *CDS Colloquium.* New York University, New York, NY, Sept. 2024.
3. Community detection in the hypergraph stochastic block model, *SINE Seminar.* University of Illinois Urbana-Champaign, Champaign, IL, Sept. 2024.
4. Community detection in the hypergraph stochastic block model, *CS Theory Seminar.* Johns Hopkins University, Baltimore, MD, Sept. 2024.
5. Community detection in the hypergraph stochastic block model and reconstruction on hypertrees, *Conference on Learning Theory (COLT) 2024.* Edmonton, CA, Canada, June 2024
6. Weak Recovery Threshold for the Hypergraph Stochastic Block Model, *Rutgers Discrete Mathematics Seminar.* Rutgers University, Piscataway, NJ, Mar. 2024
7. Reconstruction on trees and hypertrees, *Computer Science/Discrete Mathematics Seminar II.* Institute for Advanced Study, Princeton, NJ, Feb. 2024
8. Fast Sampling of b-Matchings and b-Edge Covers, *ACM-SIAM Symposium on Discrete Algorithms (SODA) 2024.* Alexandria, VA, Jan. 2024
9. Uniqueness of Belief Propagation Fixed Point, *Short Talks by Postdoctoral Members.* Institute for Advanced Study, Princeton, NJ, Oct. 2023
10. Weak Recovery Threshold for the Hypergraph Stochastic Block Model, *Probability in High Dimensions Reading Seminar.* Princeton University, Princeton, NJ, Oct. 2023
11. Weak Recovery Threshold for the Hypergraph Stochastic Block Model, *Conference on Learning Theory (COLT) 2023.* Virtual, June 2023

12. Uniqueness of BP fixed point for the Potts model and applications to community detection, *Conference on Learning Theory (COLT) 2023.* Virtual, June 2023
13. Information Theory and Uniqueness of Belief Propagation Fixed Point, *Graduate Student Probability Conference.* University of Wisconsin-Madison, Madison, WI, Sept. 2022
14. Channel degradation and uniqueness of BP fixed point beyond binary alphabet, *Conference on Information Sciences and Systems (CISS).* Virtual, Mar. 2022
15. Stochastic block model entropy and broadcasting on trees with survey, *Conference on Learning Theory (COLT) 2021.* Virtual, Aug. 2021
16. Faster monotone min-plus product, range mode, and single source replacement paths, *International Colloquium on Automata, Languages and Programming (ICALP) 2021.* Virtual, July 2021
17. Broadcasting on trees near criticality, *IEEE International Symposium on Information Theory (ISIT) 2020.* Virtual, July 2021
18. Strong data processing inequalities and reconstruction problems, *LIDS Student Conference 2020.* Massachusetts Institute of Technology, Cambridge, MA, Jan. 2020

SERVICE

– Reviewer for IEEE International Symposium on Information Theory (ISIT) 2021, 2023, 2024, 2025; IEEE Information Theory Workshop (ITW) 2022, 2023; ACM-SIAM Symposium on Discrete Algorithms (SODA) 2023, 2024; ACM Symposium on Theory of Computing (STOC) 2024, 2025; International Colloquium on Automata, Languages and Programming (ICALP) 2024; IEEE Symposium on Foundations of Computer Science (FOCS) 2024; Conference on Learning Theory (COLT) 2025; IEEE Transactions on Information Theory; Algorithmica; Annals of Statistics; Transactions on Algorithms; Journal of Computer and System Sciences; Mathematical Reviews; Bernoulli

– Program committee member for International Conference on Algorithmic Learning Theory (ALT) 2024, 2025

TEACHING

– Instructor, NYU MATH-UA 148/MA-UY 3054 Honors Linear Algebra, Spring 2025

– Instructor, NYU MA-UY 3113 Advanced Linear Algebra and Complex Variables, Fall 2024

– Teaching Assistant, MIT 6.441 Information Theory, Fall 2021

SKILLS

Programming languages: C++, Python, LaTeX

Languages: Mandarin, English, Japanese